

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n} = 1$$

$$\alpha_n = \sqrt[n]{n} - 1$$

$$n = (\alpha_n + 1)^n = 1 + n\alpha_n + \frac{1}{2}n(n-1)\alpha_n^2 + \dots$$

$$> 1 + \frac{1}{2}n(n-1)\alpha_n^2$$

← 保证 α 前次数
高于 n

$$\Rightarrow \alpha_n < \sqrt{\frac{2}{n}}$$

$$\text{取 } N = \left\lceil \frac{2}{\varepsilon^2} \right\rceil + 1$$

$$\text{当 } n > N \text{ 时 有 } |\sqrt[n]{n} - 1| \leq \sqrt{\frac{2}{n}} < \varepsilon$$

给 n 降阶

$$\sqrt[n]{n} = \sqrt[n]{\sqrt{n} \cdot \sqrt{n} \cdot \underbrace{1 \cdot \dots \cdot 1}_{n-2 \uparrow 1}} \leq \frac{2\sqrt{n} + n - 2}{n}$$

$$0 < \sqrt[n]{n} - 1 \leq \frac{2\sqrt{n} - 2}{n} = \frac{2(\sqrt{n} - 1)}{n}$$

$$\forall \varepsilon > 0 \exists N \text{ 当 } n > N, |\sqrt[n]{n} - 1| < \varepsilon$$

关键 $\alpha_n = \sqrt[n]{n} - 1$ 取等?

$$\text{若用 } \sqrt[n]{n} - 1 < \varepsilon$$

$$n < (\varepsilon + 1)^n$$

$$n < 1 + n\varepsilon + \dots \quad \text{方向反.}$$

* 错误尝试

$$n = (\alpha_n + 1)^n > 1 + n\alpha_n$$

$$\alpha_n < \frac{n-1}{n} = 1 - \frac{1}{n} < \varepsilon$$

$$n < \frac{1}{1-\varepsilon}$$

$$n = 1 + \frac{1}{2}n(n-1)\alpha_n^2$$

$$\alpha_n < \sqrt{\frac{2}{n}} < \varepsilon$$

$$n > \frac{2}{\varepsilon^2}$$

$$\lim_{n \rightarrow \infty} q^n = 0 \quad |q| < 1 \quad \text{取 } \frac{1}{1-|q|} = 1+\alpha \quad (\alpha > 0)$$

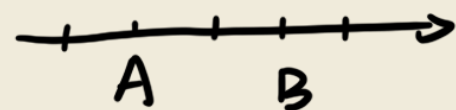
$$|q^n| = \left(\frac{1}{1+\alpha}\right)^n = \frac{1}{(1+\alpha)^n} < \frac{1}{n\alpha} < \varepsilon$$

$$\text{取 } N = \left[\frac{1}{\varepsilon\alpha}\right] + 1 \quad \text{则 } |q^n - 0| < \varepsilon, \quad \forall n > N$$

极限唯一

$$\text{设 } \lim_{n \rightarrow \infty} x_n = A \quad \lim_{n \rightarrow \infty} x_n = B$$

反证. 若 $A \neq B$ 不妨设 $A < B$



$$\text{取 } \varepsilon = \frac{B-A}{2} > 0, \text{ 由 } \lim_{n \rightarrow \infty} x_n = A, \quad \exists N_1 \in \mathbb{N}^+$$

$$\Rightarrow |x_n - A| < \varepsilon, \quad \forall n > N_1$$

$$\text{由 } \lim_{n \rightarrow \infty} x_n = B, \quad \exists N_2 \in \mathbb{N}^+$$

$$\Rightarrow |x_n - B| < \varepsilon, \quad \forall n > N_2$$

$$\text{取 } N = \max \{N_1, N_2\}$$

$$\text{当 } n > N \text{ 时 } x_n < A + \varepsilon = \frac{A+B}{2}$$

$$x_n > B - \varepsilon = \frac{A+B}{2}$$

s.t. 矛盾

从而 $A = B$

收敛数列有界

$$\{x_n\} \quad \lim_{n \rightarrow \infty} x_n = A$$

$$\text{对 } \varepsilon = 1, \text{ 由 } \lim_{n \rightarrow \infty} x_n = A, \quad \exists N \in \mathbb{N}^+ \text{ s.t. } |x_n - A| < \varepsilon = 1, \quad \forall n > N$$

$$|x_n| < |A| + 1, \quad \forall n > N$$

$$\text{取 } M = \max \{|x_1|, \dots, |x_N|, |A| + 1\}$$

$$\text{则 } |x_n| \leq M, \quad \forall n \in \mathbb{N}$$

狄利克雷函数极限不存在

假设 $\lim_{x \rightarrow a} D(x) = L$ 取 $\varepsilon = \frac{1}{2}$

$\exists \delta > 0$ s.t. $0 < |x-a| < \delta$ 时, 有 $|D(x)-L| < \frac{1}{2}$

在 a 的 δ 邻域内 $\exists r \in \mathbb{Q}, i \in \mathbb{R}$ 而 $\notin \mathbb{Q}$

$$|D(r)-L| = |1-L| < \frac{1}{2}$$

$$|D(i)-L| = |0-L| = |L| < \frac{1}{2}$$

矛盾. $\Rightarrow$ 极限不存在.

构造收敛于 a 的数列.

or. $D(r_n) \quad 1, 1, 1, \dots \quad a, a, \dots$
 $D(i_n) \quad 0, 0, 0, \dots \quad a + \frac{\sqrt{2}}{n} \text{ 逼近 } a$

$$\lim_{n \rightarrow \infty} D(r_n) = 1$$

$$\lim_{n \rightarrow \infty} D(i_n) = 0$$

$$a + \frac{\sqrt{2}}{n} \rightarrow a.$$

根据海涅归结原理. 不存在极限.

海涅归结原理(常用于证明极限不存在)

$$\lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow \forall \{x_n\} \subset \overset{\circ}{N}(x_0) \quad \lim_{n \rightarrow \infty} x_n = x_0. \quad \text{s.t.} \quad \lim_{n \rightarrow \infty} f(x_n) = A$$

$$\Rightarrow \forall \varepsilon > 0, \text{ 由 } \lim_{x \rightarrow x_0} f(x) = A \text{ 知 } \exists \delta \text{ s.t. } |f(x)-A| < \varepsilon, \forall x \in \overset{\circ}{N}_\delta(x_0)$$

$$\text{对上述 } \delta > 0, \lim_{n \rightarrow \infty} x_n = x_0, \text{ 知 } \exists N \text{ s.t. } 0 < |x_n - x_0| < \delta \quad n > N.$$

$$\text{则 } |f(x_n) - A| < \varepsilon, \forall n > N.$$

反证法

⇐ 若 $\lim_{x \rightarrow x_0} f(x) \neq A$, 则 $\exists \varepsilon_0 > 0$, 对 $\forall \delta > 0$, $\exists x_n \in \overset{\circ}{N}_\delta(x_0)$ s.t. $|f(x_n) - A| \geq \varepsilon_0$

对 $\delta = 1$, $\exists x_1, 0 < |x_1 - x_0| < 1, |f(x_1) - A| \geq \varepsilon_0$

对 $\delta = \frac{1}{2}$, $\exists x_2, 0 < |x_2 - x_0| < \frac{1}{2}, |f(x_2) - A| \geq \varepsilon_0$

对 $\delta = \frac{1}{n}$, $\exists x_n, 0 < |x_n - x_0| < \frac{1}{n}, |f(x_n) - A| \geq \varepsilon_0$

构造数列 $\{x_n\}$, $0 < |x_n - x_0| < \frac{1}{n}, \forall n \in \mathbb{N}^+$

则 $\{x_n\} \subset N(x_0)$ $\lim_{n \rightarrow \infty} x_n = x_0$

数列允许有 i 项相等, 但数列
不可能一直相等, 只需满足该数列极限趋于 x_0

② 设 $\lim_{n \rightarrow \infty} x_n = A$ (有限或 $\pm \infty$), 证 $\lim_{n \rightarrow \infty} \frac{1}{n} (x_1 + x_2 + \dots + x_n) = A$

若 A 有限 $\exists N_1$, s.t. $n > N_1$ 时 $|x_n - A| < \frac{\varepsilon}{2}$

令 $C = \sum_{k=1}^N |x_k - A|$

有 $\left| \frac{1}{n} \sum_{k=1}^n (x_k - A) \right| = \left| \frac{C}{n} + \sum_{k=N+1}^n x_k \cdot \frac{1}{n} - A \right| < \frac{C}{n} + \frac{\varepsilon}{2}$

$\exists N_2 = \left\lceil \frac{2C}{\varepsilon} \right\rceil + 1$, s.t. $n > N_2$ 时 $\frac{C}{n} < \frac{\varepsilon}{2}$

$N = \max \{N_1, N_2\}$ 当 $n > N$ 时有 $\lim_{n \rightarrow \infty} \frac{1}{n} (x_1 + x_2 + \dots + x_n) = A$

$A = +\infty$. 则 C 并非常量.

若 $A = +\infty$, $\forall M > 0$, $\exists N$ s.t. $n > N$ 时 $x_n > 2M$

$$\text{对于 } n > N, \quad \frac{1}{n} \sum_{k=1}^n x_k + \frac{1}{n} \sum_{k=N+1}^n x_k > \frac{1}{n} \sum_{k=1}^N x_k + \frac{n-N}{n} \cdot 2M$$

当 $n > 2N$ 时, 原式 $> M$

$A = -\infty$ 同理.

② 设 $x_n > 0$, $\lim_{n \rightarrow \infty} x_n = A$, 证 $\lim_{n \rightarrow \infty} \sqrt[n]{x_1 x_2 \cdots x_n} = A$

$$A > 0. \quad \text{令 } y_n = \ln x_n \quad \lim_{n \rightarrow \infty} y_n = \ln A$$

$$\text{由上题知 } \frac{1}{n} \sum_{k=1}^n \ln x_k = \ln A \Rightarrow \lim_{n \rightarrow \infty} x_1 x_2 \cdots x_n = A^n$$

$$A = 0 \quad \forall \varepsilon > 0, \exists N \text{ s.t. } n > N \text{ 时 } x_n < \varepsilon$$

$$\text{对于 } n > N, \quad x_1 x_2 \cdots x_n < x_1 x_2 \cdots x_N \cdot \varepsilon^{n-N}$$

$$\Rightarrow \sqrt[n]{x_1 x_2 \cdots x_n} < (x_1 x_2 \cdots x_N)^{\frac{1}{n}} \cdot \varepsilon^{\frac{n-N}{n}} \rightarrow \varepsilon \quad \leftarrow \text{常规证法}$$

而 ε 取任意 $\varepsilon > 0$. 故极限为 0.

取 $\ln$. 关注 $A > 0$.

正因 $A = 0$.

$|x_n - A| < \varepsilon$ 才易于使用

$$\text{若 } A = 0 \quad \text{则} \quad 0 < \sqrt[n]{x_1 x_2 \cdots x_n} \leq \frac{x_1 + x_2 + \cdots + x_n}{n}$$

$$0 \leq \lim_{n \rightarrow \infty} \sqrt[n]{x_1 x_2 \cdots x_n} \leq \lim_{n \rightarrow \infty} \frac{x_1 + x_2 + \cdots + x_n}{n} = A = 0.$$

$$\text{故} \quad \lim_{n \rightarrow \infty} \sqrt[n]{x_1 x_2 \cdots x_n} = A = 0$$

若要证

$$A = \infty. \quad \forall M > 0, \exists N_1 \in \mathbb{N}, \text{ s.t. } n > N_1 \Rightarrow x_n > 2M$$

$$x_1 \cdot x_2 \cdots x_n > x_1 \cdot x_2 \cdots x_{N_1} (2M)^{n-N_1}$$

$$\Rightarrow \sqrt[n]{x_1 \cdot x_2 \cdots x_n} > (x_1 \cdot x_2 \cdots x_{N_1})^{\frac{1}{n}} (2M)^{\frac{n-N_1}{n}}$$

$$\sqrt[n]{\frac{x_1 \cdot x_2 \cdots x_{N_1} \cdot (2M)^{N_1}}{(2M)^{N_1}} \cdot (2M)^{\frac{n-N_1}{n}}} = \left(\frac{x_1 \cdot x_2 \cdots x_{N_1}}{(2M)^{N_1}} \right)^{\frac{1}{n}} \cdot (2M)^{\frac{n-N_1}{n}}$$

$$\exists N_2, \text{ 当 } n > N_2 \text{ 时, } \frac{(x_1 \cdot x_2 \cdots x_{N_1})^{\frac{1}{n}}}{(2M)^{\frac{N_1}{n}}} > \frac{1}{2}$$

$$N = \max\{N_1, N_2\} \text{ 原式 } > M^n \Rightarrow \lim_{n \rightarrow \infty} \sqrt[n]{x_1 x_2 \cdots x_n} = A$$

极限的四则运算 前提: 存在极限 (收敛)

$$\lim_{n \rightarrow \infty} (x_n + y_n) = \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n = A + B$$

$$\alpha_n = x_n - A \quad \beta_n = y_n - B \quad (\alpha_n, \beta_n \text{ 为无穷小量})$$

$$x_n + y_n = A + \alpha_n + B + \beta_n = A + B + (\alpha_n + \beta_n)$$

$$(x_n + y_n) - (A + B) = \alpha_n + \beta_n \text{ 为无穷小量.}$$

$$\text{则 } \lim_{n \rightarrow \infty} x_n + y_n = A + B$$

$$\lim_{n \rightarrow \infty} x_n = A \Leftrightarrow \alpha_n = x_n - A \quad x_n = A + \alpha_n \quad \alpha_n (n \rightarrow +\infty) \text{ 为无穷小量}$$

$$\lim_{n \rightarrow \infty} y_n = B \Leftrightarrow \beta_n = y_n - B \quad y_n = B + \beta_n \quad \beta_n (n \rightarrow +\infty) \text{ 为无穷小量}$$

$$x_n \cdot y_n = (A + \alpha_n)(B + \beta_n) = A \cdot B + A \cdot \beta_n + B \cdot \alpha_n + \alpha_n \cdot \beta_n \quad A \cdot \beta_n, B \cdot \alpha_n, \alpha_n \cdot \beta_n \text{ 为无穷小量}$$

$$= A \cdot B.$$

$$B \neq 0 \Rightarrow \exists N \text{ s.t. } |y_n| \geq \frac{1}{2}|B| > 0 \quad \forall n > N$$

$$\left| \frac{x_n}{y_n} - \frac{A}{B} \right| = \left| \frac{Bx_n - Ay_n}{By_n} \right| = \left| \frac{B\alpha_n - A\beta_n}{B \cdot y_n} \right| \leq \frac{2}{|B|^2} |B\alpha_n - A\beta_n| \quad \forall n > N$$

常量 无穷小量

$$\lim_{n \rightarrow \infty} \left| \frac{x_n}{y_n} - \frac{A}{B} \right| = 0$$

复合函数 $\lim_{x \rightarrow x_0} f(g(x)) = A$

目标 $\forall \varepsilon > 0 \quad \exists \delta > 0 \text{ s.t. } |f(g(x)) - A| < \varepsilon \quad \forall 0 < |x - x_0| < \delta$

由 $\lim_{u \rightarrow u_0} f(u) = A$ 知 $\exists \alpha > 0 \text{ s.t. } \forall 0 < |u - u_0| < \alpha, |f(u) - A| < \varepsilon$

对上述 α , $\lim_{x \rightarrow x_0} g(x) = u_0 \quad \exists \delta > 0 \text{ s.t. } \forall 0 < |x - x_0| < \delta, |g(x) - u_0| < \alpha$

(关键) 而 $g(x) \neq u_0$ 知 $\forall 0 < |x - x_0| < \delta \quad 0 < |g(x) - u_0| < \alpha$

而 $|f(g(x)) - A| < \varepsilon \quad \forall 0 < |x - x_0| < \delta$

例: $f(u) = \begin{cases} 1 & u \neq 0 \\ 0 & u = 0 \end{cases} \quad \lim_{u \rightarrow 0} f(u) = 1$

$g(x) = x \cdot \sin \frac{1}{x} \quad \lim_{x \rightarrow 0} g(x) = 0$

$\lim_{x \rightarrow 0} f \circ g(x)$ 不存在

夹逼

$g(x) < f(x) < h(x)$

$\lim_{x \rightarrow x_0} g(x) = A \quad \lim_{x \rightarrow x_0} h(x) = A$

$\forall \varepsilon > 0, \exists \delta_1 > 0, \delta_2 > 0, \text{ s.t. } 0 < |x - x_0| < \delta_1 \text{ 时有 } |g(x) - A| < \varepsilon; 0 < |x - x_0| < \delta_2 \text{ 时, } |h(x) - A| < \varepsilon$

取 $\delta = \min \{ \delta_1, \delta_2 \}$ 当 $0 < |x - x_0| < \delta$ 时 $\left. \begin{matrix} L - \varepsilon < g(x) < L + \varepsilon \\ L - \varepsilon < h(x) < L + \varepsilon \end{matrix} \right\} \Rightarrow L - \varepsilon < f(x) < L + \varepsilon$

证 $\lim_{x \rightarrow 0} a^x = 1 (a > 0)$

$0 < x < 1 \quad \exists ! n \text{ s.t. } \frac{1}{n+1} \leq x < \frac{1}{n}$

$a^{\frac{1}{n+1}} < a^x < a^{\frac{1}{n}}$
 $x \rightarrow 0, \Rightarrow n \rightarrow \infty$

$\Rightarrow \lim_{x \rightarrow 0} a^x = 1$

此处相当于默认 $\lim_{n \rightarrow \infty} a^{\frac{1}{n}} = 1$ 已用 ε, δ 语言证明

将函数化为数列

$\{x_n\}$ 单增有上界

证: 记 $A = \sup \{x_n\}$ 确界存在公理

断言 $\lim_{n \rightarrow \infty} x_n = A$

$\forall \varepsilon > 0$ 由 $A = \sup \{x_n\}$ $A - \varepsilon$ 不再为 $\{x_n\}$ 的上界

$\exists n_0$ s.t. $x_{n_0} > A - \varepsilon$

取 $N = n_0$ 由 $\{x_n\}$ 单增知 $x_n \geq x_{n_0} > A - \varepsilon, \forall n > N$

又 $x_n \leq A, \forall n > N$ 即证 $|x_n - A| < \varepsilon, \forall n > N$

等价无穷小

$$\alpha \sim \beta \Leftrightarrow \lim \frac{\alpha}{\beta} = 1 \Leftrightarrow \alpha = \beta + o(\beta)$$

$$\Updownarrow \\ \lim \frac{\alpha - \beta}{\beta} = 0 \Leftrightarrow \alpha - \beta = o(\beta)$$

从定 $\times$ 上
 $\alpha - \beta$ 是 β 的高阶无穷小.

等价无穷小因子替换定理 (因子是乘积)

$$\alpha \sim \alpha_1, \beta \sim \beta_1$$

$$\lim (\alpha \cdot \beta) = \lim (\alpha_1 \cdot \beta_1)$$

$$\lim \frac{\alpha \cdot \beta}{\gamma \cdot \delta} = \lim \frac{\alpha_1 \cdot \beta_1}{\gamma_1 \cdot \delta_1}$$

对于 $\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3}$ 强行用加减法 $\lim_{x \rightarrow 0} \frac{x + o(x) - [x + o(x)]}{x^3}$

泰勒展式 0 阶. 不精确

